

Def. Define

$$f: [0,1] \rightarrow \mathbb{R}; x \mapsto \begin{cases} \frac{1}{p} & \text{if } x \in \mathbb{Q} \text{ with } x = \frac{q}{p}, \text{ irreducible form,} \\ 0 & \text{otherwise.} \end{cases}$$

Then,

- 1) f is continuous at every irrational numbers, and discontinuous at every rational numbers.
- 2) f is Riemann-integrable.

Proof. observe that for each $n \in \mathbb{N}$, the set

$$S_n = \{x \in [0,1] \mid f(x) > \frac{1}{n}\} = f^{-1}\left[\left(\frac{1}{n}, \infty\right)\right] = \left\{\frac{q}{p} \in [0,1] \mid \gcd(p,q)=1, 1 \leq p \leq n\right\}.$$

is finite set,

$$\min\{x_0 - y \mid y \in S_n\}.$$

Let $x_0 \in [0,1]$ be an irrational number.

Given $\varepsilon > 0$, there is $N \in \mathbb{N}$ s.t. $\frac{1}{N} < \varepsilon$. Since the S_N is finite, we can take $\delta > 0$ s.t.

$$x \in (x_0 - \delta, x_0 + \delta) \Rightarrow |f(x_0) - f(x)| = f(x) \leq \frac{1}{N} < \varepsilon.$$

So, f is continuous at any irrational. And, for rational $\frac{q}{p} \in [0,1]$,

for any $\delta > 0$, there is an irrational $x \in (\frac{q}{p} - \delta, \frac{q}{p} + \delta)$ s.t. $|\frac{1}{p} - f(x)| = \frac{1}{p} > \frac{1}{2p}$.

So it is not continuous at rational. $\square$

Given $\varepsilon > 0$, let $N \in \mathbb{N}$ be a number such that $\frac{1}{N} < \frac{\varepsilon}{2}$.

Denote that $S_N = \{x_1, \dots, x_k\}$. Since this S_N is finite, put $d > 0$ such that

$$d < \min\{|x_i - x_j| \mid i \neq j\} \quad \text{and} \quad d \cdot k < \frac{\varepsilon}{2}.$$

Now, construct a partition on $[0,1]$ as

$$P = \left\{0, 1, x_1 - \frac{d}{2}, x_1 + \frac{d}{2}, \dots, x_k - \frac{d}{2}, x_k + \frac{d}{2}\right\} \cap [0,1].$$

Then, by the density of irrationals, the lower sum $\underline{L}(f, P) = 0$, and

$$\mathcal{U}(f, P) \leq 1 \cdot \frac{1}{N} + 1 \cdot d \cdot k$$

$$\text{Now, } \int f - \int f \leq \mathcal{U}(f, P) - \underline{L}(f, P) \leq \frac{1}{N} + d \cdot k < \frac{\varepsilon}{2} + \frac{\varepsilon}{2}$$

gives the result, by the integrable criterion. $\square$